Karishma Deepak Galande

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Summary

Software Engineer with over 5 years at Bank of America writing C#, Java, and SQL for regulated financial systems. Pursuing a Master of Engineering in Computer Science at the University of Cincinnati with a focus on databases, distributed systems, query performance and reliable data services.

Education

University of Cincinnati, College of Engineering and Applied Science *Cincinnati, Ohio*
Master of Engineering, Computer Science Expected: April 2028
Relevant Coursework: Databases, Distributed Systems, Software Testing & Quality Assurance, AI, ML

MKSSS's Cummins College of Engineering for Women *Pune, India*
Bachelor of Technology, Computer Engineering May 2021
Relevant Coursework: Data Structures and Algorithms, Object Oriented Programming, Databases, Big Data

Technical Skills

Languages & Web: Java, C#, Python, SQL, C++, JavaScript, TypeScript, React, Angular, Node.js, Flask
Databases: SQL Server, DB2, Oracle, MySQL, CockroachDB, MongoDB (NoSQL), FAISS
Backend & Cloud: Spring Boot, .NET Core, ASP.NET MVC, AWS (Lambda, Bedrock, S3), Azure, Docker, OpenShift
DevOps & Quality: Git, Bitbucket, Jira, Jenkins CI/CD, Ansible, SonarQube, Checkmarx, SDLC, Code Review, xUnit, Agile
AI & ML: GitHub Copilot (AI assisted coding), PyTorch, TensorFlow, LLMs, RAG, LangChain, Hugging Face

Experience

University of Cincinnati, Digital Technology Solutions *Cincinnati, Ohio*
Student Associate | Software Engineering Sep 2026 - Present

- Debug and test ASP.NET applications for College of Medicine admissions, course selection, and course lottery systems, tracing defects through application logs and SQL data.
- Build SQL and Excel test datasets covering edge cases to validate form logic and lottery results before release.
- Create Power BI dashboards for monitoring usage and defect trends across 4 course lottery selection processes.
- Write technical reference documents guiding students through each application form before submission.

Bank of America *Mumbai, India*
Software Engineer 1A | Full-Stack Development Jan 2026 - Jul 2026

- Tuned 100+ SQL queries and stored procedures per application, cutting DB2 data retrieval time 30% with zero downtime.
- Reviewed pull requests in Bitbucket and wrote xUnit tests for .NET services, verifying AI generated code from GitHub Copilot before merge.
- Decomposed 6+ ASP.NET MVC monoliths into Java Spring Boot microservices on OpenShift, removing 4 redundant service dependencies across 3 engineering teams.
- Migrated 3 web apps and 3 web services from ASP.NET MVC to .NET Core for cloud hosting, cutting deployment time 25%.
- Built an LLM report generation tool on the bank's internal AI platform with human review before release, cutting turnaround from 4 or 5 days to under 30 minutes for hundreds of advisors.
- Resolved 1,500+ SonarQube and Checkmarx findings across 8 application portfolios, reducing security vulnerabilities 40%.

Software Engineer 1B | Full-Stack Development Jun 2021 - Dec 2025

- Replaced licensed ASPOSE with open-source NPOI in 2 sprints, saving $30,000 a year with zero report regressions.
- Automated 6 manual deployment steps with Jenkins CI/CD, moving 2 teams from monthly to weekly releases.
- Converted 3 synchronous workflows to async REST APIs across 5 high traffic tools, cutting page load time 75%.
- Implemented Content Security Policy and HTTPS across 6 production applications, cutting vulnerability exposure from over 80% to under 5%.

- Troubleshot on call production issues across 8 applications using logs and SQL, shipping fixes within 12 hours.

Projects

Bank Support Agent | *Python*, *LangGraph*, *Amazon Bedrock*, *AWS Lambda*, *MCP*

- Built a LangGraph agent on Amazon Bedrock with 4 tools for policy RAG, transaction lookup, interest calculation, and disputes, deployed on AWS Lambda and exposed through an MCP server.
- Added guardrails for prompt injection and PII redaction, validated by a 20 case evaluation set covering tool selection, accuracy, and latency.

Clinical RAG for Drug Interaction Detection | *Python*, *LangChain*, *Hugging Face*, *FAISS*, *Docker*, *React*

- Built a RAG pipeline with Hugging Face embeddings and FAISS vector retrieval over 50,000 public clinical records, detecting drug interactions at 0.91 F1.
- Served the pipeline through a Dockerized REST API with JWT role based access control and a React Redux front end.

Multimodal Assistive Caption Bot | *Python*, *TensorFlow*, *Flask*, *NLP*, *Deep Learning*

- Fine tuned a CNN encoder and RNN decoder in TensorFlow on 10,000+ images and paired it with text to speech, turning an image into a spoken caption in under 3 seconds for visually impaired users.

Drowsiness and Fatigue Detection System | *Java*, *OpenCV*, *Image Processing*

- Deployed a real time OpenCV fatigue detection model on a Raspberry Pi with no server dependency, detecting eye closure, yawning, and head movement at 95% accuracy with under 200 ms inference latency.

Certifications

- AI for Engineers: Core Concepts | Bank of America AI Pathway, Degreed
- AI for Engineers: Awareness Curriculum | Bank of America AI Pathway, Degreed
- React JS | Bank of America Certification
- Angular Programming | Bank of America Certification